Pricing and hedging financial derivatives
Exercise class 1

Exercise 1 (Properties of Calls and Puts in the absence of arbitrage). Let $\mathrm{Call}_t(T,K)$ (respectively $\mathrm{Put}_t(T,K)$) be the price at date t of a European Call (Put) option with strike K and expiry date T, on a non-dividend paying asset whose price is denoted by S. We assume that there are no arbitrage opportunities. The price of a zero-coupon bond with maturity T will be denoted by $B_t(T)$ and we suppose that $0 < B_t(T) \leq 1$.

1. Show that

$$(S_t - KB_t(T))^+ \leq \mathrm{Call}_t(T,K) \leq S_t \quad \text{and} \quad (KB_t(T) - S_t)^+ \leq \mathrm{Put}_t(T,K) \leq KB_t(T).$$

2. Show that $\mathrm{Call}_t(T,K)$ is decreasing in K and $\mathrm{Put}_t(T,K)$ is increasing in K.
3. Show that $\mathrm{Call}_t(T,K)$ and $\mathrm{Put}_t(T,K)$ are convex in K.

We suppose from now on that the risk-free rate r is positive and constant and we can trade the risk free asset $S^0_t = e^{rt}$.

4. Show that for $\Theta \geq 0$, and $0 \leq t \leq T$,

$$\mathrm{Call}_t(T+\Theta,K) \geq \mathrm{Call}_t(T,Ke^{-r\Theta}) \quad \mathrm{Put}_t(T+\Theta,K) \geq \mathrm{Put}_t(T,Ke^{-r\Theta}).$$

5. Show that $\mathrm{Call}_t(T,K)$ is increasing in T. Discuss the case of a Put.

Exercise 2 (Collar option). We consider a financial instrument with pay-off

$$\min(\max(S_T,K_1),K_2), \quad K_1 < K_2.$$

1. Plot the pay-off of this instrument as function of S_T.
2. Decompose the price at date t into a combination of zero-coupons and call options.

Exercise 3 (Quadratic hedging). Consider an arbitrage-free financial market where trading takes place only at $t = 0$ and $t = T$. In this market, one can buy/sell a risk-free asset, with zero interest rate, and a forward contract on a risky asset, whose initial value is S_0 and terminal value is denoted by S_T. Our goal is to find the best way to hedge an option with pay-off $g(S_T)$ at time T by investing into the risk-free asset and the forward contract. We assume that S_T and $g(S_T)$ are square integrable random variables.

1. We define the quadratic hedging strategy for the pay-off $g(S_T)$ as the portfolio which minimizes the L^2 distance to the option pay-off at maturity:

$$\min_{\alpha,\beta} \mathbb{E}\left[(g(S_T) - \alpha - \beta(S_T - S_0))^2\right].$$

   Compute the coefficients α and β.

2. Compute the price of the quadratic hedging strategy at time $t = 0$.

3. Show that if one defines the price of the option as the price of its quadratic hedging strategy, such a pricing rule is linear, preserves the constants and respects the put-call parity. Show that a simple pricing rule by expectation ($\Pi(X) = \mathbb{E}[X]$) may not respect the put-call parity.

4. Application to a specific model where $S_t = S_0 + \mu t + \sigma W_t$, where W is a Brownian motion.

Exercise 4 (Chooser option). Assume an arbitrage-free market, with constant risk-free rate r. We are interested in the so called *chooser* option, which allows the holder to choose, at date T_0, between a call option with maturity $T > T_0$ and strike K and a put with the same maturity / strike.

1. Argue why the price of this option at time T_0 is given by

$$\mathrm{Chooser}_{T_0}(T, K) = \max(\mathrm{Call}_{T_0}(T, K), \mathrm{Put}_{T_0}(T, K)).$$

2. What is the terminal pay-off of this option at time T?

3. Deduce from the put-call parity that the price at time T of the chooser option may be decomposed as follows:

$$\mathrm{Chooser}_{T_0}(T, K) = \mathrm{Call}_{T_0}(T, K) + V_{T_0}(T, K),$$

   where $V_{T_0}(T, K)$ is a cash flow to be determined.

4. Compute the price of this option at time $t \in [0, T_0]$ and give an exact hedging strategy using vanilla options.

Exercise 5 (Option strategies). Draw the pay-off of the following strategies at time T as function of S_T and identify what are the expectations about market behavior implied by each strategy.

- Sale of a stock, purchase of two calls with strike K (straddle).
- Purchase of a call and a put with the same strike K (this is also called a straddle, why?)
- Purchase of a call and two puts with strike K (strip).
- Purchase of two calls and a put with strike K (strap).
- Purchase of a call with strike K_1 and a put with strike K_2 (strangle).
- Purchase of a call with strike K_1 and sale of a call with strike $K_2 > K_1$ (bull spread).
- Purchase of a call with strike K_2 and sale of a call with strike $K_1 < K_2$ (bear spread).
- Purchase of a call with strike K_1, sale of two calls with strike $(K_1 + K_2)/2$ and purchase of a call with strike K_2, with $K_1 < K_2$ (butterfly spread).